

NICOLE LEE

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EDUCATION

University of California, Berkeley

Expected Graduation: May 2028

B.S. in Electrical Engineering and Computer Science, B.S. in Bioengineering

- **Hong Kong Scholarship for Excellence Scheme Scholar** - USD\$155,000 scholarship to study at UC Berkeley
- **Coursework:** Data Structures & Algorithms, Computer Networks, Web Development, Computer Architecture, Machine Learning, Databases, Intractable Algorithms, Internet Architecture, User Design, Computational Biology
- **Activities:** Web Development at Berkeley, Cal Hacks (Director), Notion Campus Leader, Society for Women Engineers

SKILLS

Languages: Typescript, C/C++, Java, Python, Swift, Objective-C, Go, Rust, SQL, RISC-V

Frontend: SwiftUI, UIKit, AppKit, Xcode, XCTest, Linux, Git, Bash, CI/CD, Distributed Systems, React

Backend: FastAPI/REST API, Node.js, Express.js, PostgreSQL, Redis, MongoDB, AWS (EC2, S3, Lambda), Azure, Docker, Kafka

AI/ML: TensorFlow, PyTorch, MediaPipe, Hugging Face, YOLOv12

EXPERIENCE

Software Engineer Intern

May 2026 - Aug 2026

Apple

Cupertino, CA

- **Built a full MCP Evaluation platform** across **450+** servers, saving **\$700,000+/year** by generating **200+** **schema-aware** ground truth queries per run and automating live tool calls to cut **per-eval time 18x** (6+hrs → 20mins)
- **Architected fault-tolerant tracing proxy** intercepting live MCP traffic with response normalization and persistent PostgreSQL state, collapsed **100+** **correlated failures into <10 root causes**, cutting **triage 40%**.
- **Engineered hybrid scoring** with deterministic tool-call/JSON diffing + LLM-as-judges; extracted Langfuse traces + A/B runs to analyze routing failures and recommend tool description/routing changes optimizing accuracy, latency & cost

Forward Deployed Engineer Intern

Aug 2025 – Dec 2025

Wedge (YC S25)

San Francisco, CA

- Built a clinician analytics platform for LA General Medical Center (600-bed Level I trauma center) using React, TypeScript, and GraphQL, surfacing EHR and scheduling across **30K+** **annual discharges** to improve bed turnover.
- Architected **20+** **reusable frontend components** and **REST data pipelines** for FHIR/HL7 EHR data, adding client-side caching, conflict detection, and utilization scoring to support low-latency analytics under concurrent workloads.

Software Engineer Intern

Jun 2025 – Aug 2025

Anchor Logics

Berkeley, CA

- **Led 5 engineers** to ship a real-time telemetry platform that analyzes **3D IMU data streamed from IoT vests** used by **50+** **ALS/Parkinson's patients**, surfacing data visualizations, timeline scrubbers, and playback controls.
- Built CV and data infrastructure with YOLOv12, OpenPose, DeepSORT, AWS S3, Redis, and PostgreSQL, improving **keypoint consistency 25%**, cutting occlusion loss 35%, and supporting **2,000+** **models with sub-100ms retrieval**.

Director

Aug 2024 – Present

Cal Hacks [Link]

Berkeley, CA

- **Built and deployed 3 production websites** including live hacker portal, judging system, and public site (React, Typescript, Supabase, PostgreSQL) and published iOS app (C, Swift) for the largest collegiate hackathon, serving **4,000+** **hackers and 60+** **sponsors**.
- Built an **automated sponsorship pipeline** reaching **2,000+** **companies** and **14,000+** **contacts**, raising **\$1,250,000+**.

Software Engineer Intern

Apr 2025 – Jun 2025

Digpath.ai

Berkeley, CA

- **Deployed a SageMaker + Meta SAM cell segmentation pipeline** with GPU-accelerated inference across 500+ whole-slide pathology images (70GB dataset); **reduced manual annotation** turnaround from days to under **2 hours** per case.
- **Optimized serverless platform** (AWS Amplify, Lambda, DynamoDB) tracking pathology image ingestion, storage and case volume.

PROJECTS

CodeMRI [Link] | Python · TypeScript · Tree-sitter · FastAPI

- Built a **static-analysis + context compiler for coding agents**, parsing repository-wide codebase ASTs into persistent architecture and symbol graphs with call/import dependencies, source navigation, and change-impact analysis
- Compiled change-scoped context **reducing repository-reading tokens by 83%** and enabling graph-guided code modifications.

Flusk (Flush + Disk LSM Tree) [Link] | Java 17 · Maven · Guava · Snappy · JUnit 5 · Railway

- Built a Java **LSM-tree storage engine** with full write path (WAL, MemTable, SSTable, 4-tier compaction with tombstone GC, Bloom filters and sparse indexing); exposed 7 REST endpoints for low-latency reads/writes

Stoich [Link] | Next.js · TypeScript · MongoDB Atlas · RDKit-JS · 3Dmol.js · D3.js

- **Built a chemistry research tool** via multi-LLM pipeline and cross-paper vector search, reducing atom-index hallucinations from **67% → <15%**; renders synchronized reaction graphs, equation and mechanism views, annotations, and 3D molecular structures.